$\Rightarrow$ every time for same x_0 we get diff ϵ_t $\forall t$

there are mutiple x_t for x_0

$\Rightarrow$ There doesn't exist a single (a) a unique x_t the generates a given x_0, from the reverse diffusion process

$\Rightarrow$ This is non unique invertibility of DDPM

we donot have unique latent vector to generate an image.

* The above issues of slow sampling & non-unique invertability of DDPMs are solved with DDIMs.

DDIMs:

Let us consider a family of inference distributions: parameterized by a scalar $\sigma \in \mathbb{R} \geq 0$

$q_\sigma(x_{1:T} \mid x_0)$ posterior dist of latent given data

$$q_\sigma(x_{1:T} \mid x_0) = q_\sigma(x_T \mid x_0) \prod_{t=2}^{T} q_\sigma(x_{t-1} \mid x_t, x_0)$$

where

$$q_\sigma(x_T \mid x_0) \triangleq N(\sqrt{\alpha_T}\, x_0, (1-\alpha_T) I) \quad \text{and}$$

$\forall\ t > 1$ This is the choice of the model

$$q_\sigma(x_{t-1} \mid x_t, x_0) \triangleq N\left(\sqrt{\sigma_{t-1}}\, x_0 + \sqrt{1-\alpha_{t-1} - \sigma_t^2}\ \frac{x_t - \sqrt{\alpha_t}\, x_0}{\sqrt{1-\alpha_t}},\ \sigma_t^2 I\right)$$

The above process is a non Markovian.

$$q_\sigma(x_t \mid x_{t-1}, x_0) = \frac{q_\sigma(x_{t-1} \mid x_t, x_0)\; q_\sigma(x_t \mid x_0)}{q_\sigma(x_{t-1} \mid x_0)}$$

The forward process, corresponding to above posterior is as follows.

The above forward process is not markovian.

$$x_t \not\perp\!\!\!\perp x_0 \mid x_{t-1}$$

* In the above process

$$q_\sigma(x_t \mid x_0) = N\left(x_t ; \sqrt{\alpha_t}\, x_0, \sqrt{1-\alpha_t} \cdot I\right)$$

This is same as the DDPM.

($\bar{\alpha}_t$ in DDPM is same as α_t in DDIM)

* Since $q_\sigma(x_t \mid x_0)$, conditional of x_t given x_0 is the same in both DDPM & DDIM, The ELBO for both of them are the same & optimizing the ELBO leads to the same solu^n

$\therefore$ the ELBO in DDPM or DDIM depends only on $q(x_t \mid x_0)$

$\Rightarrow$ as long as $q(x_t | x_0)$ is same as that of DDPMs, any process can be used and leads to the same soln for ELBO

as far as $q(x_t | x_0)$ is same as DDPM, any defn for $q(x_{1:T} | x_0)$ can be choosen.

* No parameters in forward process.

Defining the Decoder (a) the reverse process:

$$p_\theta(x_{t-1} | x_t) = \begin{cases} N(\mu_\theta(x_1), \sigma^2 I) & t = 1 \\ q_\sigma(x_{t-1} | x_t, \mu_\theta(x_t)) & \text{others} \end{cases}$$

where $$\mu_\theta(x_t) = \frac{x_t - \sqrt{1-\alpha_t}\,\hat{\epsilon}_\theta}{\sqrt{\alpha_t}}$$

Recall $$x_t = \sqrt{\alpha_t}\, x_0 + \sqrt{1-\alpha_t} \cdot \epsilon_t, \quad \epsilon_t \sim N(0, I)$$

$$x_0 = \frac{x_t - \sqrt{1-\alpha_t} \cdot \epsilon_t}{\sqrt{\alpha_t}}$$

Recall the consistency term in DDPM

$$D_{KL}(p_\theta(x_{t-1} | x_t) \,\|\, q(x_{t-1} | x_t, x_0))$$

here x_0 is approximated by $\mu_\theta(x_t)$

The main result is the ELBO corresponding to the above non-markovian family of models & that of DDPM are the same (off by a const)

$\Rightarrow$ Training the Regression n/w in DDPM is implicitly training all the above non-markovian models.

$\Rightarrow$ Training process for DDPM & DDIM remains same where as the diff lies in inference, as the reverse process are not same.

$\Rightarrow$ while we train one DDPM, this gets inf DDIM family

$\Rightarrow$ DDIM samping will take less time, compared to DDPM.

Inference with DDIMs:

* Sampling from $p_\theta(x_{t-1}|x_t)$

$$p_\theta^{\cdot}(x_{t-1}|x_t) = \begin{cases} \mathcal{N}(\mu_{\theta}^{\cdot}(x_1), \sigma^2 I) & t=1 \\ q_\sigma(x_{t-1}|x_t, \mu_\theta^{\cdot}(x_t)) & \text{others} \end{cases}$$

$$q_\sigma(x_{t-1} \mid x_t, x_0) \doteq N\left(\sqrt{\sigma_{t-1}}\,\mu_{\theta^*} + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\;\frac{x_t - \sqrt{\alpha_t}\,\mu_{\theta^*}}{\sqrt{1-\alpha_t}},\ \sigma_t^2 I\right)$$

where $$\mu_{\theta^*}(x_t) = \frac{x_t - \sqrt{1-\alpha_t}\;\hat{\epsilon}_{\theta^*}}{\sqrt{\alpha_t}} \quad \text{— (A)}$$

Recall $$x_t = \sqrt{\alpha_t}\,x_0 + \sqrt{1-\alpha_t}\cdot\epsilon_t, \quad \epsilon_t \sim N(0, I)$$

$$x_0 = \frac{x_t - \sqrt{1-\alpha_t}\cdot\epsilon_t}{\sqrt{\alpha_t}}$$

putting all the above together, Sampling from $p_{\theta^*}(x_{t-1} \mid x_t)$ is as follows:

$$x_{t-1} = \sqrt{\alpha_{t-1}}\,\underbrace{\left(\frac{x_t - \sqrt{1-\alpha_t}\cdot\hat{\epsilon}_{\theta^*}}{\sqrt{\alpha_t}}\right)}_{\mu_{\theta^*}} + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\;\underbrace{\hat{\epsilon}_{\theta^*}}_{\text{obtained by rearrange (A)}} + \sigma_t^2 z$$

$z \sim N(0, I)$

when $$\sigma_t = \sqrt{\frac{(1-\alpha_{t-1})}{\sqrt{1-\alpha_t}}}\;\frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_{t-1}}}$$

The above random process reduces to DDPM.

$\Rightarrow$ DDPM is a special case of DDIM.

* When $\sigma_t = 0$, the sampling process has the following form.

$$x_{t-1} = \sqrt{\alpha_{t-1}} \left[\frac{x_t - \sqrt{1-\alpha_t} \cdot \hat{\epsilon}_{\theta^*}}{\sqrt{\alpha_t}} \right] + \sqrt{1-\alpha_{t-1}} \cdot \epsilon_{\theta^*}$$

The above sampling process is deterministic.

$\Rightarrow$ The only randomness is from $x_T \sim N(0, I)$

With $\sigma_t = 0$, the above model is called DDIM.

$\Rightarrow$ The forward process in DDIM with $\sigma_t = 0$, is also deterministic.

The above implies the DDIM has a unique invertibility (DDIM Inversion)